

Final Project

Introducció a l'Estadística 2026

Analysis of Weight Factors

UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH

Facultat d'Informàtica de Barcelona



Iván Soriano
Ariel Litrán
Joel Nicolás
Nil Muriach

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1. Introduction

Obesity is currently one of the most important public health issues, since it is associated with a higher risk of several chronic diseases and other health complications. For this reason, it is interesting to study which lifestyle and demographic factors may be related to body weight.

In this project, we analyze a dataset of 2,111 individuals from Mexico, Peru and Colombia (such is the description of the dataset, but the data is artificial) (AravindPCoder, 2023). The dataset includes 17 variables related to demographic characteristics, eating habits, physical condition and obesity level, which makes it suitable for descriptive, inferential and predictive analysis (AravindPCoder, 2023, this is the reference to the dataset, the link is in the Bibliography section).

The main objective of this study is to examine how these variables are associated with body weight and other obesity-related outcomes, and to evaluate whether weight can be predicted through a multiple linear regression model. To do this, the analysis is divided into three main parts: descriptive analysis, bivariate inference and predictive modeling, in this case a multiple regression model.

More specifically, we study whether daily water consumption is associated with weight, whether the type of transport used is related to physical activity, and whether dietary and physical-activity-related variables help explain differences in body weight.

To ensure a more precise and rigorous study, this project focuses explicitly on body weight as a continuous numerical measurement, both in the analysis of the data and in our goals, not taking into account categorical obesity classifications that could limit statistical detail.

Overall, the aim of this project is to provide a clear statistical analysis of obesity-related factors using R and focusing on weight, combining interpretation of the data with predictive modeling tools.

2. Descriptive Analysis

2.1 Summary

Since we already did an in-depth descriptive analysis in the last submission, we opted to summarize this part in order to make the report more digestible.

Overall, the dataset gives us a broad and quite complete picture of the sample, since it includes 2,111 individuals and no missing values in the variables considered. The sample is also very balanced by gender, while the age distribution shows that most participants are relatively young, with a mean age of 24.31 years.

In terms of physical characteristics, height is fairly concentrated around the mean, whereas weight shows much greater variability, which already suggests noticeable differences between individuals in the dataset. Looking at lifestyle-related variables, several patterns stand out: most participants report a family history of overweight, frequent consumption of high-calorie food, and occasional snacking between meals. At the same time, smoking and calorie monitoring are clearly uncommon behaviors in this sample, while public transportation is by far the most common means of transport. Finally, the obesity-level categories are distributed across all seven groups, with Obesity Type I being the most frequent, which suggests that the dataset contains substantial variation and is suitable for deeper descriptive, inferential, and predictive analyses.

Here are two summary tables to better visualize this study:

Variable	Type	N	Missing	Summary
Gender	Categorical	2111	0	Levels = 2; Mode = Male (1068, 50.59%)
Age	Numeric	2111	0	Mean = 24.31; SD = 6.35; Median = 22.78; Min = 14; Q1 = 19.95; Q3 = 26; Max = 61
Height	Numeric	2111	0	Mean = 1.70; SD = 0.09; Median = 1.70; Min = 1.45; Q1 = 1.63; Q3 = 1.77; Max = 1.98
Weight	Numeric	2111	0	Mean = 86.59; SD = 26.19; Median = 83; Min = 39; Q1 = 65.47; Q3 = 107.43; Max = 173

family_history _with_overwei ght	Categorical	2111	0	Levels = 2; Mode = yes (1726, 81.76%)
FAVC	Categorical	2111	0	Levels = 2; Mode = yes (1866, 88.39%)
FCVC	Numeric	2111	0	Mean = 2.42; SD = 0.53; Median = 2.39; Min = 1; Q1 = 2; Q3 = 3; Max = 3
NCP	Numeric	2111	0	Mean = 2.69; SD = 0.78; Median = 3; Min = 1; Q1 = 2.66; Q3 = 3; Max = 4
CAEC	Categorical	2111	0	Levels = 4; Mode = Sometimes (1765, 83.61%)
SMOKE	Categorical	2111	0	Levels = 2; Mode = no (2067, 97.92%)
CH2O	Numeric	2111	0	Mean = 2.01; SD = 0.61; Median = 2; Min = 1; Q1 = 1.58; Q3 = 2.48; Max = 3
SCC	Categorical	2111	0	Levels = 2; Mode = no (2015, 95.45%)
FAF	Numeric	2111	0	Mean = 1.01; SD = 0.85; Median = 1; Min = 0; Q1 = 0.12; Q3 = 1.67; Max = 3
TUE	Numeric	2111	0	Mean = 0.66; SD = 0.61; Median = 0.63; Min = 0; Q1 = 0; Q3 = 1; Max = 2
CALC	Categorical	2111	0	Levels = 4; Mode = Sometimes (1401, 66.37%)
MTRANS	Categorical	2111	0	Levels = 5; Mode = Public_Transportation (1580, 74.85%)
NObesyedad	Categorical	2111	0	Levels = 7; Mode = Obesity_Type_I (351, 16.63%)

Table 1: General descriptive summary of the variables included in the obesity dataset

Variable	Category	Count	Percent
Gender	Female	1043	49.41
Gender	Male	1068	50.59
family_history_with_overweight	no	385	18.24
family_history_with_overweight	yes	1726	81.76
FAVC	no	245	11.61
FAVC	yes	1866	88.39
CAEC	Always	53	2.51
CAEC	Frequently	242	11.46
CAEC	no	51	2.42
CAEC	Sometimes	1765	83.61
SMOKE	no	2067	97.92
SMOKE	yes	44	2.08
SCC	no	2015	95.45
SCC	yes	96	4.55
CALC	Always	1	0.05
CALC	Frequently	70	3.32
CALC	no	639	30.27
CALC	Sometimes	1401	66.37
MTRANS	Automobile	457	21.65
MTRANS	Bike	7	0.33

MTRANS	Motorbike	11	0.52
MTRANS	Public_Transportation	1580	74.85
MTRANS	Walking	56	2.65
NObeyesdad	Insufficient_Weight	272	12.88
NObeyesdad	Normal_Weight	287	13.60
NObeyesdad	Obesity_Type_I	351	16.63
NObeyesdad	Obesity_Type_II	297	14.07
NObeyesdad	Obesity_Type_III	324	15.35
NObeyesdad	Overweight_Level_I	290	13.74
NObeyesdad	Overweight_Level_II	290	13.74

Table 2 : Frequency distribution of the categorical variables in the obesity dataset.

Additionally, we did include the R scripts of the in-depth descriptive analysis in the appendix part.

If for some reason you do not have access to the last submission, which includes the mentioned study in detail, here it is:

<https://drive.google.com/file/d/1B1V16ZWb2F3jLvGL7dI9bEkCGqKcdF5/view?usp=sharing>

3. Description of the choice of the inference techniques

3.1 Elimination and preparation of variables

When we were analyzing the different variables in the dataset, we found the following data regarding the variables SMOKE and SCC (calorie tracking):

Smoke		SCC	
no	yes	no	yes
2067	44	2015	96
97.91568%	2.08432%	95.452392 %	4.547608%

Table 3: Distribution of Smoking and SCC

After analyzing the variables SMOKE and SCC (calorie tracking), the results clearly show a significant imbalance between the number of people who answered "no" versus those who answered "yes". Only 2.08% of the sample smokes, and a mere 4.54% track their calories. Due to this evident disparity in the data for these two variables, we decided to exclude them from the main analyses to avoid inaccurate results.

3.2 Bivariate Inference

Association between daily water consumption and weight

To study whether daily water consumption is associated with body weight, we analyzed the relationship between CH2O and Weight using correlation analysis. Since both variables are quantitative, correlation is an appropriate method to evaluate the strength and direction of their association. Pearson's correlation was used as the main measure because it assesses linear association between two numerical variables. In addition, Spearman's correlation was computed as a complementary non-parametric measure in order to confirm whether the relationship was also consistent in terms of monotonic association.

To examine whether daily water consumption is associated with body weight, we analyzed the relationship between CH2O and Weight using a scatterplot and correlation tests. Since both variables are quantitative, correlation analysis is an appropriate method to evaluate the

strength and direction of their association. We chose Pearson's correlation as the main measure because it evaluates the linear association between two numerical variables, which fits our goal.

However, the CH2O variable is discrete and only takes values from 1 to 3, so it clearly doesn't follow a normal continuous distribution. Because our sample size is very large (2,111 individuals), the Central Limit Theorem actually compensates for this, meaning the Pearson p-value is still perfectly valid. Even so, we wanted to double-check our results to be completely sure. After a bit of research, we decided to run Spearman's correlation as a backup because it is a non-parametric test that works with data rankings. It doesn't require normality at all, so it helped us confirm that the positive trend we found is solid and not just caused by the limited 1 to 3 scale (UCLA, 2020).

To do such analysis we are going to compute the results using R (Statistics Bio7, 2025).

Association between means of transport and physical activity

To evaluate the relation between physical activity (FAF) and the means of transport (MTRANS) we did a Chi-squared test. Although the variable FAF presents continuous values, in other words, has values with decimals (probably because some data is synthetic), which means it's a quantitative variable (Chi-squared requires both variables to be qualitative), we rounded the values to their nearest integer so that we get 4 separated groups. This results in the variable becoming qualitative, which lets us do the test mentioned. The groups are:

Level of physical activity:

- 0: None.
- 1: 1-2 days a week
- 2: 2-4 days a week
- 3: 4-5 days a week

The null hypothesis is that the means of transport and the frequency of physical activity are independent, and we use a significance level of 0.05.

Due to the fact that for doing the Chi-squared test we modified one of the variables, which could result in invalid results, we will also conduct another test (which is not part of the theory seen in class) that doesn't require changing anything from the variables, and then we

will compare them to see if the conclusions match. Lastly, we will also represent the data of the variables in boxplots, so that we can see it more easily (this will be done before the tests).

To do so, we need to choose between using an ANOVA analysis of variance and a Kruskal-Wallis test (which are the two options when we have 1 quantitative variable and the other one qualitative).

To determine if an ANOVA analysis of variance could be used to study the relationship between the transport used and physical activity, it was first necessary to verify that the quantitative variable (FAF) followed a normal distribution. To do this, we performed a normal Q-Q plot of the residuals.

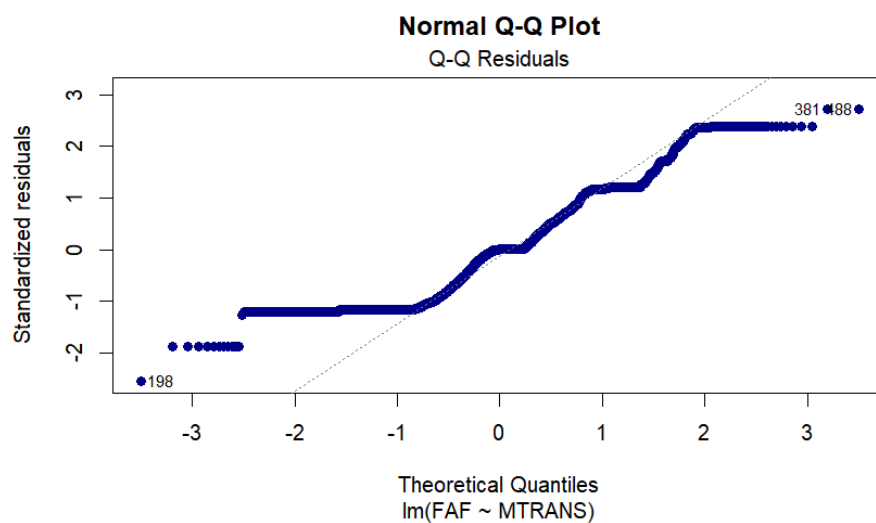


Figure 1: Q-Q Plot Transport/physical activity

The plot displayed a severe deviation from the diagonal reference line, showing a distinct step-like pattern and significant asymmetry. This clear deviation tells us that it doesn't follow a normal distribution, which means we discard the ANOVA analysis and will instead perform the Kruskal-Wallis test and a Wilcoxon post-hoc test, an ideal test for comparing data that do not follow a normal distribution (Prabhakaran, 2026).

3.3 Predictive Model: Multiple Linear Regression

We chose multiple linear regression because this technique allows us to evaluate the combined impact of various lifestyle and demographic factors on an individual's weight at the same time. By using this model, we can isolate the individual effect of each predictor and, for example, compare the influence of dietary habits versus the influence of physical activity while holding all other variables constant. Furthermore, it lets us know how precisely the variables of our model can predict a person's weight.

We left “NObeyesdad” out of our predictor variables to prevent data leakage, as a person's obesity category is derived directly from their weight.

Before training the model, we randomly split the dataset: 80% for training the algorithm and 20% for testing its accuracy on new data and preventing the model from just memorizing our sample (Temitope, 2024).

To make sure that our model is reliable, we made a few standard quality checks. First, we used a correlation matrix to visually see the correlation between two variables and check that two variables aren't giving the same information. After that we used the Variance Inflation Factor (VIF) to confirm that our variables weren't providing redundant information in a more precise way (Vikas, 2024).

Furthermore, we made a normal Q-Q plot of the residuals to confirm that our model's overall error follows a normal distribution to confirm that our model isn't biased. Once we confirmed the model was mathematically right, we evaluated its actual performance using the Adjusted R-squared (to see how well it explains the data), alongside the Mean Absolute Error (MAE) which is used to measure the average prediction error in kilograms, and the Root Mean Squared Error (RMSE), which is used to penalize more severe prediction mistakes (Juan, 2022) .

4. Results of the analysis of the data

4.1 Bivariate Inference results

Association between daily water consumption and weight

After the R script execution, these were the found results:

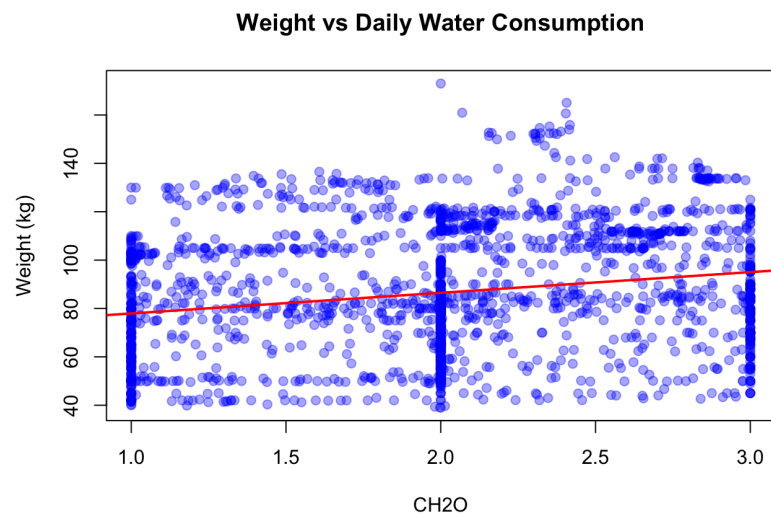


Figure 2 : Scatterplot of CH2O and Weight, with fitted linear regression line.

The scatterplot shows a slight upward trend, suggesting that individuals with higher daily water consumption tend to have somewhat higher weight values. However, the points are widely dispersed, so the relationship does not appear to be strong. Since CH2O is measured on a limited scale from 1 to 3, the observations are concentrated in vertical bands, which is expected for this variable.

Test	Estimate	95% CI	p-value	Interpretation
Pearson correlation	0.2006	[0.1593, 0.2412]	$< 2.2e-16$	Weak positive linear association.
Spearman correlation	0.2256	—	$< 2.2e-16$	Weak positive monotonic association.

Table 4. Pearson and Spearman correlation results for the association between daily water consumption (CH2O) and body weight (Weight).

As shown in Table 4, Pearson's correlation coefficient was 0.2006, with a 95% confidence interval of [0.1593, 0.2412] and a p-value below $2.2e-16$. This shows a statistically significant positive association between CH2O and Weight, although the strength of the relationship is weak. Spearman's correlation gave a very similar result, with a coefficient of 0.2256 and a p-value below $2.2e-16$. Overall, both tests point to the same conclusion: daily water consumption and body weight are positively associated in this sample, but the association is not strong.

In conclusion, regarding the bivariate analysis, daily water consumption showed a weak but statistically significant positive association with body weight.

Association between means of Transport and physical activity

The boxplots we got were these ones:

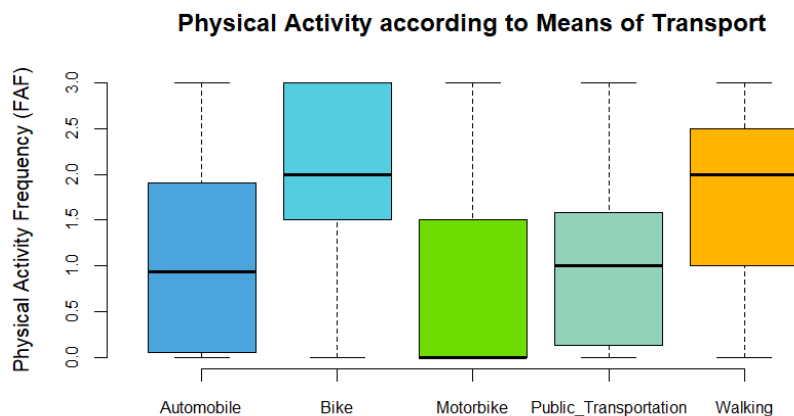


Figure 3: Physical Activity according to means of Transport

We can see that people who walk or cycle tend to have higher levels of physical activity than those who do not, such as those who use cars or public transport.

Next, we did the contingency table and the Chi-squared test, with these results:

Group	0	1	2	3
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Automobile	162	149	116	30
Bike	1	1	2	3
Motorbike	7	1	1	2
Public_Transportation	539	609	362	70
Walking	11	16	15	14

Table 5: Contingency table

X-squared = 79.611	df = 12	p-value = 4.897e-12
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Table 6: Chi-squared test

The Chi-squared test for independence has as results a test statistic of $X\text{-squared} = 79.611$ with 12 degrees of freedom and a p-value of 4.897×10^{-12} . Since the p-value is significantly smaller than our significance level (0.05), we reject the null hypothesis, which means that there is strong statistical evidence to conclude that an individual's choice of transport is associated with their frequency of weekly physical activity.

Although we rejected the null hypothesis, it's important to remark on the fact that if we look at the expected values of the Chi-squared test, we find 2 values lower than 1 (in theory there shouldn't be any) and 45% of the values are lower than 5 (ideally it should be 20% or less), which means that the result could be invalid. These values are due to the fact that Bike and Motorbike categories have extremely low samples, so low that it causes these results. Despite this approximation limitation, the fact that the p-value we got is so much smaller than 0.05 strongly confirms that the association is highly substantial and not an artifact of the test's distribution choice.

Now that we already have the results of the Chi-squared test, we will do the other alternative that we mentioned earlier: the Kruskal-Wallis test and a Wilcoxon post-hoc test.

Kruskal-Wallis test:

Kruskal-Wallis chi-squared = 23.52	df = 4	p-value = 9.969e-05
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Table 7:Kruskal-Wallis test

As the p-value is much smaller than 0.05 ($0.00009969 < 0.05$) we can reject the null hypothesis, as we know there is a difference in physical activity depending on the transport you use, but we don't know how much difference there is between each option. That's why we need to do a post-hoc test. As the distribution is not normal (as we saw earlier), the best option is doing a Wilcoxon post-hoc test.

Wilcoxon post-hoc test:

	Automobile	Bike	Motorbike	Public_Transportation
Bike	0.19814	-	-	-
Motorbike	1.00000	0.67765	-	-
Public_Transportation	1.00000	0.14720	1.00000	-
Walking	0.00227	1.00000	0.39950	0.00073

Table 8:Wilcoxon post-hoc test

By analyzing the results of the Wilcoxon post-hoc test, we can reject the initial hypothesis for p-values less than 0.05, such as 0.00073 or 0.00227. In these cases, we reject the initial hypothesis and conclude that the two categories do indeed have a significant difference in the amount of physical activity performed. On the other hand, p-values greater than 0.05 (or 1) indicate that there is no difference in the amount of physical activity performed between the groups or that the initial hypothesis cannot be refuted with sufficient certainty.

For example, we can see a difference between Walking/Automobile and Walking/Public_Transportation, and if we look at the boxplots, we see that Walking is, on average, higher than Automobile and Public_Transportation. On the other hand, the Bike group doesn't have a single p-value lower than 0.05, even though the boxplots show it's well above the average of the other options. This is because the sample of people in the Bike category is so small that the test doesn't dare to conclude that the level of physical activity is different.

Comparison of the 2 tests:

After comparing both tests (Chi-squared test and the Kruskal-Wallis) we can see they both get similar results. The Chi-squared test on rounded FAF levels showed a highly significant global dependency (with $p\text{-value} = 4.897 \times 10^{-12}$), though it triggered an approximation warning because 45% of expected cells were below 5 (and also 2 values were below 1), which was due to the small sample size of Bike (Sample of 7) and Motorbike (Sample of 11).

To verify that the rounded values did not produce invalid results, the Kruskal-Wallis test used the FAF variable without any modification (as a continuous variable), which robustly confirmed the significant global dependency (with $p\text{-value} = 9.969 \times 10^{-5}$). Additionally, the Wilcoxon post-hoc test showed that this significance is strictly driven by the contrasts between Walking/Automobile ($p\text{-value} = 0.00227$) and Walking/Public_Transportation ($p\text{-value} = 0.00073$). Both Bike and Motorbike categories lacked sufficient statistical power to show significant differences with other groups (as mentioned earlier), which resulted in the Bike group not having any $p\text{-value}$ lower than 0.05 despite the boxplots showing that it was higher than the other groups. The fact that both tests conclude there is a difference in physical activity depending on the transport used confirms that the modified values of FAF did not introduce any directional bias into our study.

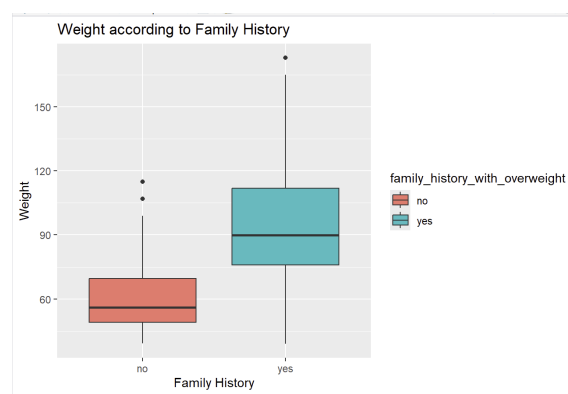


Figure 4: Boxplot of Weight according to family history with overweight.

The boxplot shows that individuals with a family history of overweight tend to present higher body weights than individuals without family history.

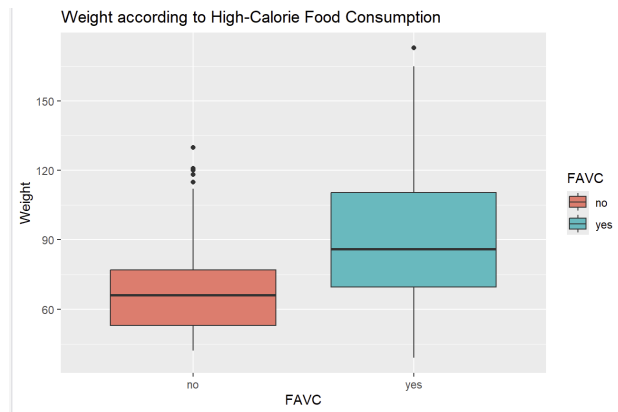


Figure 5: Boxplot of Weight according to FAVC.

Individuals who frequently consume high-calorie food tend to present higher body weights.

4.2 Multiple Linear Regression results

Before the regression model, we created a correlation matrix to see the relationship between the numerical variables.

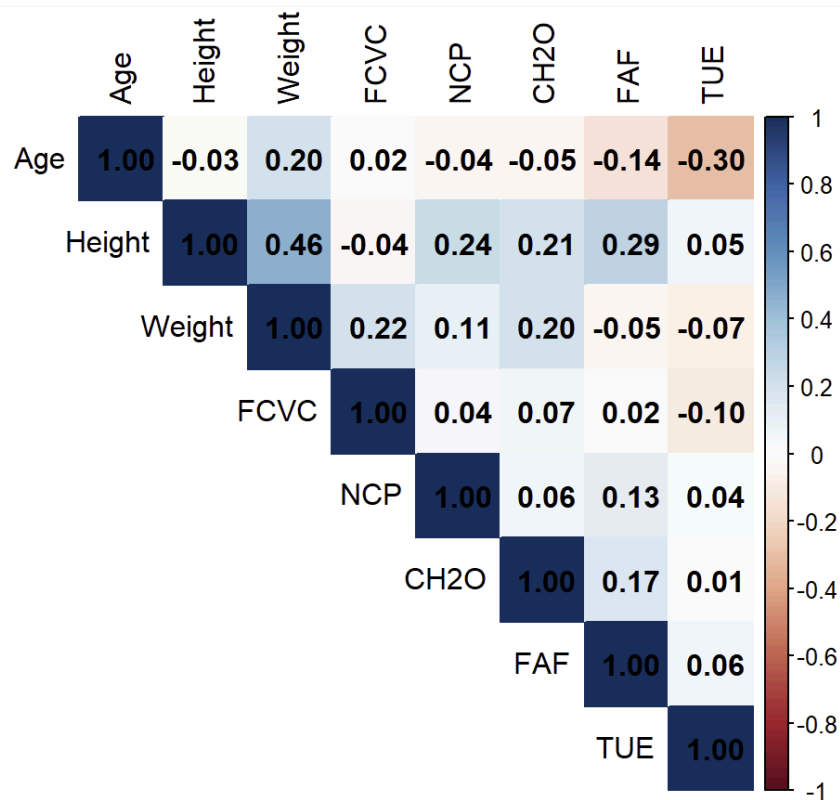


Figure 6: Correlation matrix of numerical variables.

The matrix shows that most correlations are low or moderate. The strongest relationship is between Height and Weight. This makes sense, because taller people usually tend to weigh more. There are no very high correlations between the predictors, so there is no strong evidence of serious multicollinearity among them.

Predictor	Estimate	Std. Error	t value	p-value	95% Conf. Interval	Sig.
(Intercept)	-204.65	20.53	-9.967	< 0.001	[-244.92, -164.38]	***
Gender (Male)	-5.42	1.17	-4.626	< 0.001	[-7.72, -3.12]	***
Age	1.03	0.09	11.338	< 0.001	[0.85, 1.21]	***
Height	128.59	6.68	19.257	< 0.001	[115.49, 141.69]	***
Family History (Yes)	18.53	1.24	14.896	< 0.001	[16.09, 20.98]	***
FAVC (High Calorie)	4.58	1.41	3.248	0.001	[1.82, 7.35]	**
FCVC (Vegetables)	8.27	0.84	9.837	< 0.001	[6.62, 9.91]	***
NCP (Number of Meals)	0.32	0.58	0.560	0.575	[-0.81, 1.45]	
CAEC (Frequently)	-8.40	2.81	-2.995	0.003	[-13.91, -2.90]	**
CAEC (No)	9.74	3.84	2.538	0.011	[2.21, 17.27]	*
CAEC (Sometimes)	10.93	2.58	4.237	< 0.001	[5.87, 15.99]	***
CH2O (Water)	1.79	0.73	2.457	0.014	[0.36, 3.22]	*
FAF (Physical Act.)	-1.72	0.55	-3.141	0.002	[-2.79, -0.65]	**
TUE (Tech Devices)	-1.70	0.75	-2.275	0.023	[-3.17, -0.23]	*
CALC (Frequently)	-13.86	17.52	-0.791	0.429	[-48.22, 20.50]	
CALC (No)	-17.07	17.39	-0.981	0.327	[-51.19, 17.05]	
CALC (Sometimes)	-8.97	17.41	-0.515	0.607	[-43.11, 25.18]	
MTRANS (Bike)	5.80	6.60	0.879	0.380	[-7.15, 18.75]	

MTRANS (Motorbike)	16.25	5.57	2.920	0.004	[5.34, 27.17]	**
MTRANS (Public Tr.)	16.17	1.34	12.105	< 0.001	[13.55, 18.79]	***
MTRANS (Walking)	4.86	2.95	1.650	0.099	[-0.92, 10.64]	

Table 9: Multiple Linear Regression Results

“***” = p-value < 0.001 “**” = p-value < 0.01 “*” = p-value < 0.05 no “*” = p-value > 0.05

Model accuracy:

Overall, our model does a solid job of capturing the data, explaining 58.5% of the differences in people's weight (Adjusted R-squared = 0.5852). The predictors worked well together without providing redundant information, as all VIF values were under the standard threshold of 5. When we made the model do the test using 20% validation data, its predictions missed the real weight by an average of 13.57 kg MAE, with an RMSE of 16.99 kg. Considering the natural (wide variations in human body weight) this gives us a reasonable and realistic base for our predictions.

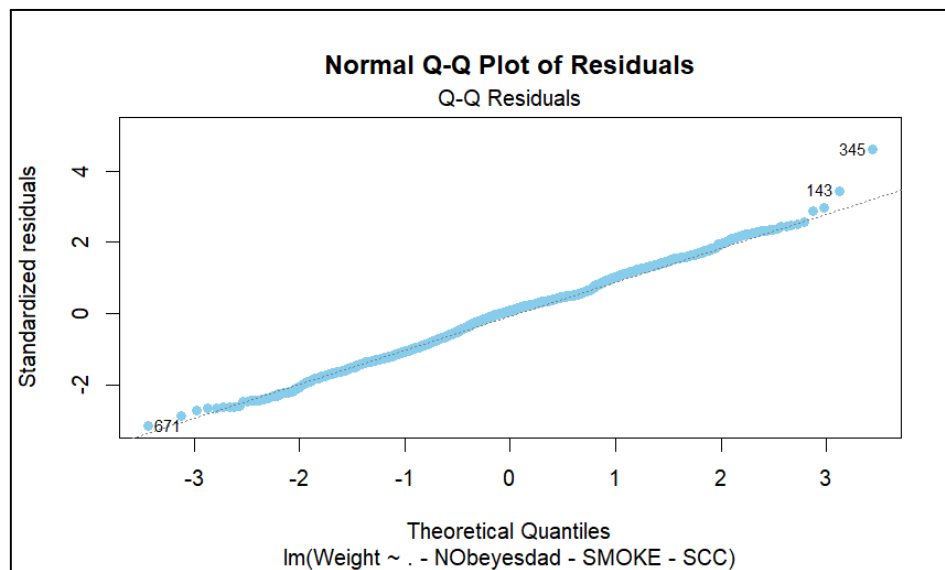


Figure 7: Normal q-q plot of the residuals

Also, as we can see in this Q-Q plot, the residual (the overall model error) follows a normal distribution, meaning that our statistical results are valid as the model's errors are random and balanced.

Interpretation of predictors and confidence intervals:

As we can see by looking at the individual variables in the table, basic demographic factors like Height, Age, and Family History of overweight are highly reliable predictors ($p < 0.001$), and also FCVC (Vegetables), MTRANS (Public Transport) and CAEC are, and we know these results are solid because their 95% confidence intervals never cross zero. For example, the interval for Age is $[0.85, 1.21]$, which means we can be 95% confident that, assuming all other habits remain the same, within the range of our sample (14 years to 61 years), every year of age adds between 0.85 kg and 1.21 kg to a person's weight.

On the other hand, some variables didn't show a clear impact. Factors like alcohol consumption (CALC), the number of main meals (NCP), using bikes (MTRANS $\rightarrow$ Bike) or walking (MTRANS $\rightarrow$ Walking) as a method of transport are unreliable predictors in our model. Because their confidence intervals include zero, their effect on weight in the wider population could be nothing.

5. Conclusions and discussion

This study shows that weight is associated with various factors, such as eating habits, physical activity, transportation, and personal characteristics such as age, height, and family history.

Regarding daily water consumption (CH_2O) and weight, as shown in the bivariate inference section, CH_2O and Weight had a positive and statistically significant association, although the relationship was weak. Therefore, people with higher water consumption tend to have higher weight. However, the correlation was weak, so CH_2O does not explain a large part of the variation in Weight.

We analyzed the relationship between the type of transport used and physical activity and found considerable variation among the different modes of transport. People who walk or use a bike tend to show higher physical activity, while those who travel by car or public transportation tend to show lower physical activity. To check whether these differences were statistically significant or could be due to chance, we performed the Kruskal-Wallis test. The Kruskal-Wallis test produced a very small p-value, indicating significant differences in FAF between some transport groups. However, Kruskal-Wallis only shows that general differences exist; it does not specify exactly which groups are different. For this reason, we also performed a Wilcoxon post-hoc test. The clearest differences were observed between walking and using a car, and between walking and using public transportation.

The results for CH_2O vs. Weight and MTRANS vs. FAF were obtained from section 4.1, Bivariate Inference Results. The predictive model results were obtained from section 4.2, Multiple Linear Regression Results, in which we concluded that the model showed a moderate but reasonable performance with the data: Adjusted $R^2 = 58.5\%$, MAE = 13.57 kg, and RMSE = 16.99 kg.

Therefore, we can conclude that the type of transport is associated with the level of physical activity. However, small categories such as Bike should be interpreted with caution. On the other hand, we also analyzed other important factors such as Height, Age, family history, FAVC, and CAEC. We found that FAVC and CAEC were significant, and FAF and TUE were also significant, but diet seems to have greater predictive power.

Finally, we can say that association does not imply causation. The dataset is partially synthetic, meaning that part of the sample was not composed of real people but artificially generated observations. In addition, we cannot guarantee that these findings apply equally to all countries.

5.1 Future improvements

Although we've conducted an initial general analysis of the topic, there are many aspects that can be improved. For example, the dataset is limited because some of the data is synthetic, meaning it's not real, which prevents us from generalizing.

Furthermore, to improve the sample, we could add data from more countries, age groups, and types of people to make the results more representative. We've also encountered the problem that some variables, such as bike, motorbike, and smoke, have very small sample sizes, which makes the results less reliable.

To conduct a more thorough analysis, we could add more variables (finding a bigger and broader dataset or combining datasets) such as income, education, sleep patterns, pre-existing conditions, actual calorie consumption, and more detailed daily habits.

To conclude with, we could also test more advanced models and compare them with multiple linear regression. Even without these improvements, the model is useful for examining the relationship between weight and lifestyle habits.

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7. Appendix (R scripts)

Descriptive Analysis

```
# Load data
datos <- read.csv("ObesityDataSet.csv", stringsAsFactors = TRUE)

# General summary
summary(datos)

# Separation of variables
numeric_data <- datos[, c("Age", "Height", "Weight", "FCVC", "NCP", "CH2O", "FAF", "TUE")]
categ_data <- datos[, c("Gender", "family_history_with_overweight", "FAVC", "CAEC",
"SMOKE", "SCC", "MTRANS", "NObeyesdad")]

# numeric variables summary
num_summary <- data.frame(
  Variable = names(numeric_data),
  Mean = round(sapply(numeric_data, mean, na.rm = TRUE), 4),
  SD = round(sapply(numeric_data, sd, na.rm = TRUE), 4),
  Min = round(sapply(numeric_data, min, na.rm = TRUE), 4),
  Q1 = round(sapply(numeric_data, quantile, 0.25, na.rm = TRUE), 4),
  Median = round(sapply(numeric_data, median, na.rm = TRUE), 4),
  Q3 = round(sapply(numeric_data, quantile, 0.75, na.rm = TRUE), 4),
  Max = round(sapply(numeric_data, max, na.rm = TRUE), 4)
)

print(num_summary)

# Correlation matrix
library(corrplot) # install.packages("corrplot")
cor_matrix <- cor(numeric_data)
corrplot(cor_matrix, method = "color", type = "upper", addCoef.col = "black", tl.col = "black")

par(mfrow = c(2, 2)) # arrange plots in a 2x2 grid

# Distributions
# Gender
barplot(table(categ_data$Gender), col = c("pink", "lightblue"),
  main = "Gender Distribution", ylab = "Count")

# Age
hist(numeric_data$Age, col = "steelblue", border = "white",
  main = "Distribution of Age", xlab = "Age (Years)", ylab = "Count")

# Weight
hist(numeric_data$Weight, col = "coral", border = "white",
  main = "Distribution of Weight", xlab = "Weight (kg)", ylab = "Count")

# Boxplot of height
```

```
boxplot(numeric_data$Height, col = "lightgreen",
        main = "Boxplot of Height", ylab = "Height (m)")

par(mfrow = c(2, 2))

# Family history
plot(categ_data$family_history_with_overweight, main="Family History of Overweight",
     col="lightblue", ylab="Frequency")

# High caloric food (FAVC)
plot(categ_data$FAVC, main="Frequent consumption of high caloric food",
     col="salmon", ylab="Frequency")

# Vegetables consumption (FCVC)
hist(numeric_data$FCVC, main="Frequency of consumption of vegetables",
     xlab="Level (1-3)", col="seagreen", breaks=5)

# number of main meals (NCP)
boxplot(numeric_data$NCP, main="Number of main meals (NCP)",
        col="skyblue", ylab="Quantity")
par(mfrow = c(2, 2))

# Daily water consumption (CH2O)
hist(numeric_data$CH2O, breaks = 20, col = "steelblue", border = "white",
     main = "Daily Water Consumption (CH2O)", xlab = "Scale (1-3)", ylab = "Frequency")

# Physical activity frequency (FAF)
hist(numeric_data$FAF, breaks = 20, col = "darkorange", border = "white",
     main = "Physical Activity Frequency (FAF)", xlab = "Scale (0-3)", ylab = "Frequency")

#snacking (CAEC)
barplot(table(categ_data$CAEC), col = "steelblue",
        main = "Food Consumption Between Meals (CAEC)", las = 2)

# Calorie monitoring (SCC)
barplot(table(categ_data$SCC), col = c("tomato", "steelblue"),
        main = "Calorie Monitoring (SCC)")
```

Bivariate inference - Association between daily consumption of water and weight

```
df <- read.csv("ObesityDataSet.csv")

str(df[, c("CH2O", "Weight")])
summary(df$CH2O)
summary(df$Weight)

# Scatterplot
```

```
plot(df$CH2O, df$Weight,
     main = "Weight vs Daily Water Consumption",
     xlab = "CH2O",
     ylab = "Weight (kg)",
     pch = 19, col = rgb(0, 0, 1, 0.35))
abline(lm(Weight ~ CH2O, data = df), col = "red", lwd = 2)

# Correlations
pearson_res <- cor.test(df$CH2O, df$Weight, method = "pearson")
spearman_res <- cor.test(df$CH2O, df$Weight, method = "spearman", exact = FALSE)

pearson_res
spearman_res
```

Bivariate inference - Association between means of Transport and physical activity

```
datos <- read.csv(file.choose(), stringsAsFactors = TRUE)

#See SMOKE and SCC information
table(datos$SMOKE)
prop.table(table(datos$SMOKE)) * 100

table(datos$SCC)
prop.table(table(datos$SCC)) * 100

# Boxplot format
par(mar=c(6.5, 4, 3, 1))

#Boxplot
boxplot(FAF ~ MTRANS, data = datos,
       main = "Physical Activity according to Means of Transport",
       xlab = "",
       ylab = "Physical Activity Frequency (FAF)",
       col = c("#4EA8DE", "#56CFE1", "#70E000", "#94D2BD", "#FFB703"),
       cex.axis = 0.82,
       frame.plot = FALSE)

# Q-Q Plot
modelo_norm <- lm(FAF ~ MTRANS, data = datos)

plot(modelo_norm, which = 2, pch = 16, col = "darkblue", main = "Normal Q-Q Plot")

#Kruskal-Wallis
kruskal.test(FAF ~ MTRANS, data = datos)

#Wilcoxon post-hoc
pairwise.wilcox.test(datos$FAF, datos$MTRANS, p.adjust.method = "bonferroni")
```

```
# Round variable
datos$FAF_discrete <- round(datos$FAF)
datos$FAF_discrete <- as.factor(datos$FAF_discrete)

#Contingency table
tabla_contingencia <- table(datos$MTRANS, datos$FAF_discrete)
print(tabla_contingencia)

#Chi-squared
chi_res <- chisq.test(tabla_contingencia, correct = FALSE)
print(chi_res)

#To see the expected values
chi_res$expected
```

Predictive Analysis - Multiple Linear Regression (Lourdes, 2021) (Juan, 2022)

```
# Predictive analysis - multiple linear regression (80/20)
# Predictive analysis - predicting weight

datos <- read.csv("ObesityDataSet.csv")

#Train/validation data split (80% / 20%)

set.seed(21)
n_rows <- nrow(datos)
train_indices <- sample(1:n_rows, size = 0.8 * n_rows)

train_data <- datos[train_indices, ]
test_data <- datos[-train_indices, ]

# Train the linear regression model

model_of_weight <- lm(Weight ~ . - NObeyesdad - SMOKE - SCC, data = train_data)

#Model summary and multicollinearity check
summary(model_of_weight)

library(car)
```

```
print("VIF Values:")
vif(model_of_weight)

# Make predictions on test data
predictions <- predict(model_of_weight, newdata = test_data)

# Calculate prediction errors (RMSE and MAE)

errors <- test_data$Weight - predictions

rmse <- sqrt(mean(errors^2))
mae <- mean(abs(errors))

cat("RMSE:", round(rmse, 2), "kg\n")
cat("MAE:", round(mae, 2), "kg\n")

# Confidence intervals and residual diagnostics

confint(model_of_weight)

par(mfrow = c(1, 1))
plot(model_of_weight, which = 2, col = "skyblue", pch = 16,
     main = "Normal Q-Q Plot of the Residuals")
```